

## Menu-Driven Program for Decreasing Patterns

### Aim

To write a menu-driven C program using a switch statement and nested loops to print different decreasing patterns based on the user's choice.

### Requirements

C compiler (GCC), text editor, and Linux/Kali terminal.

### Algorithm

1. Start the program.
2. Display a menu containing four decreasing-pattern choices.
3. Read the user's menu choice.
4. Read the number of rows n.
5. Use a switch statement to select the required pattern.
6. Use nested loops to print the selected decreasing pattern from n elements down to 1 element.
7. If the choice is invalid, display an invalid-choice message.
8. Stop the program.

### Pattern Logic

Choice 1: Print decreasing stars.

Choice 2: Print the current row number repeatedly.

Choice 3: Print the same decreasing number repeatedly in each row.

Choice 4: Print descending numbers in each row.

### C Program

```
#include <stdio.h>

int main()
{
    int choice, n, i, j;

    printf("1. Decreasing Star Pattern\n");
    printf("2. Decreasing Row-Number Pattern\n");
    printf("3. Decreasing Repeated-Number Pattern\n");
    printf("4. Decreasing Descending-Number Pattern\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    printf("Enter number of rows: ");
```

```

scanf("%d", &n);

switch (choice)
{
    case 1:
        for (i = n; i >= 1; i--) {
            for (j = 1; j <= i; j++) printf("* ");
            printf("\n");
        }
        break;

    case 2:
        for (i = n; i >= 1; i--) {
            for (j = 1; j <= i; j++) printf("%d ", i);
            printf("\n");
        }
        break;

    case 3:
        for (i = n; i >= 1; i--) {
            for (j = 1; j <= i; j++) printf("%d ", n - i + 1);
            printf("\n");
        }
        break;

    case 4:
        for (i = n; i >= 1; i--) {
            for (j = i; j >= 1; j--) printf("%d ", j);
            printf("\n");
        }
        break;

    default:
        printf("Invalid choice!\n");
}
return 0;
}

```

## Compilation and Execution

```

gcc hdoc25.c -o hdoc25
./hdoc25

```

## Test Cases

| Test | Choice | Rows | Expected Output                      |
|------|--------|------|--------------------------------------|
| 1    | 1      | 4    | Decreasing * pattern                 |
| 2    | 2      | 4    | Decreasing row-number pattern        |
| 3    | 3      | 4    | Decreasing repeated-number pattern   |
| 4    | 4      | 4    | Decreasing descending-number pattern |

## Execution Screenshots

```
(kali㉿kali)-[~/Desktop/apple]
$ gcc hdoc25.c -o hdoc25

(kali㉿kali)-[~/Desktop/apple]
$
```

*Compilation of hdoc25.c completed successfully.*

```
— Decreasing Pattern Menu —
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: 1
Enter number of rows: 4
* * * *
* * *
* *
*
```

*Choice 1 - Decreasing right-angled triangle (Rows = 4).*

```
— Decreasing Pattern Menu —
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: 2
Enter number of rows: 4
1 2 3 4
1 2 3
1 2
1
```

*Choice 2 - Decreasing natural-number pattern (Rows = 4).*

```
— Decreasing Pattern Menu —  
1. Decreasing right-angled triangle  
2. Decreasing natural-number pattern  
3. Decreasing repeated-number pattern  
4. Reverse-number triangle  
5. Exit  
Enter your choice: 3  
Enter number of rows: 4  
4 4 4 4  
3 3 3  
2 2  
1
```

*Choice 3 - Decreasing repeated-number pattern (Rows = 4).*

```
— Decreasing Pattern Menu —  
1. Decreasing right-angled triangle  
2. Decreasing natural-number pattern  
3. Decreasing repeated-number pattern  
4. Reverse-number triangle  
5. Exit  
Enter your choice: 4  
Enter number of rows: 4  
4 3 2 1  
3 2 1  
2 1  
1
```

*Choice 4 - Reverse-number triangle (Rows = 4).*

```
— Decreasing Pattern Menu —  
1. Decreasing right-angled triangle  
2. Decreasing natural-number pattern  
3. Decreasing repeated-number pattern  
4. Reverse-number triangle  
5. Exit  
Enter your choice: 5  
Exiting program.
```

```
(kali@kali)-[~/Desktop/apple]  
$ ./hdoc25
```

*Choice 5 - Exit option.*

```
— Decreasing Pattern Menu —  
1. Decreasing right-angled triangle  
2. Decreasing natural-number pattern  
3. Decreasing repeated-number pattern  
4. Reverse-number triangle  
5. Exit  
Enter your choice: 9  
Invalid choice. Please enter 1 to 5.
```

*Invalid choice handling (Choice = 9).*

## Result

The menu-driven C program was implemented successfully to generate the required decreasing patterns using a switch statement and nested loops.